🚚 Delivery Route Planner Workflow

This document outlines the 5-step pipeline to process raw order data and generate an optimized delivery route map.

⚡ Quick Start Checklist

Run these commands in order:

1. `python dataset_prep.py`
2. `python geodata.py`
3. `python preprocess_order_history.py`
4. `python build_matrix_with_distance.py` (Note: Slow processing)
5. `python run_hybrid_solver_layers.py`

🗺️ Execution Pipeline Details

1. Dataset Preparation

- **Command:** `python dataset_prep.py`
- **Input:** `order_history_kaggle_data.csv` (Required prerequisite)
- **Output:** `locations_to_geocode.csv`
- **Purpose:** Extracts unique delivery addresses from raw historical data.

2. Geocoding

- **Command:** `python geodata.py`
- **Input:** `locations_to_geocode.csv`
- **Output:** `geocoded_locations.csv`
- **Purpose:** Converts text addresses into $latitude/longitude$ coordinates for system processing.

3. Order Preprocessing

- **Command:** `python preprocess_order_history.py`
- **Input:** `geocoded_locations.csv`
- **Output:** `preprocessed_orders.csv`
- **Purpose:** Finalizes and cleans the active order list for the solver.

4. Distance Matrix Generation

- **Command:** `python build_matrix_with_distance.py`
- **Input:** `geocoded_locations.csv`

- **Outputs:**
  - `matrix_data_with_distance.json` (Travel time/distance lookup)
  - `distance_cache.json` (API/calculation cache)
- **Note:** This step is computationally intensive and may take significant time depending on the dataset size.

5. Route Optimization (Solver)

- **Command:** `python run_hybrid_solver_layers.py`
- **Inputs:**
  - `preprocessed_orders.csv` (From Step 3)
  - `matrix_data_with_distance.json` (From Step 4)
- **Dependencies:** `optimization_solver_layers.py` , `hybrid_solver_layers.py`
- **Output:** `output.html`
- **Purpose:** Executes the hybrid layered solver and generates the interactive visual route map.

Open `output.html` in any web browser to view the final results.